

PART 1: Answer the following questions briefly. Your explanations should be brief and to the point (10 %)

1. How does Java achieve platform independence?
 2. Mention two of the shortcomings of procedural programming paradigms? How does Object Oriented Programming paradigm solve these problems?
 3. What is a static member?
 4. Mention two of the advantages of inheritance.
 5. Why can't a superclass object be used to access fields and methods in its subclass?
- not a concept -> static class
bytecode
JVM*

PART 2: Show the Java implementation of the following class description. (20%)

Define a class called Currency that is used to store the representation of a money amount in any currency type (Ethiopian Birr, British Pound, ...).

- The class must have the following attributes (or properties) and their associated **accessors** and **mutators**:
 - amount (type double): the amount of money
 - symbol (type String): the symbol used when displaying that currency. For example, the symbol for US Dollars is \$, the symbol for British pounds is £, and the symbol for Ethiopian Birr is ETB.
 - usdConversionRate (type double): the rate used to convert the amount of money to U.S. Dollars.
- In addition to the accessors and mutators, the class must also have these additional methods:
 - getUSDValue: it takes no arguments and returns the amount converted to U.S. Dollars. You must use the conversion rate (usdConversionRate) for your calculation
 - displayAmount: it takes no arguments and prints the amount of money preceded by the currency's symbol (E.g. ETB100, \$200, £300 ...)
- Include one constructor in your class definition. This constructor must take the values for the amount, symbol and usdConversionRate as parameters and initialize them. ✓
- Demonstrate the usage of your class by writing a program that **converts** and **displays** 20 Ethiopian Birr (ETB) into equivalent U.S. Dollars (\$). You may use a conversion rate of 0.056 (1/18).